# Lab Report: Training dynamics of deep models

## 1 Introduction

In this report we compare several models on standard benchmarks. Each experiment was repeated 3 times with different random seeds. We trained each model for 20 epochs with a batch size of 64.

## 2 Results

Figure 1 shows the f1 score by dataset.

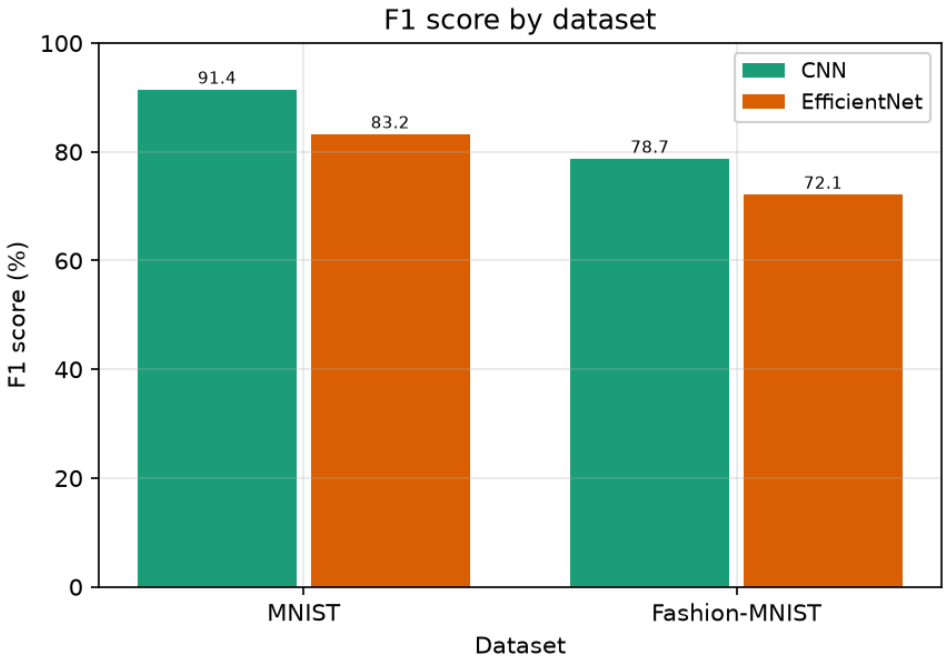



Figure 1: F1 score by dataset for the models in this experiment.

On MNIST, CNN obtains a F1 score of 93.4%. On Fashion-MNIST, CNN is about 7 points better than EfficientNet. CNN reaches 68.7% F1 score on Fashion-MNIST (Figure 1).

Figure 2 shows the validation accuracy during training.

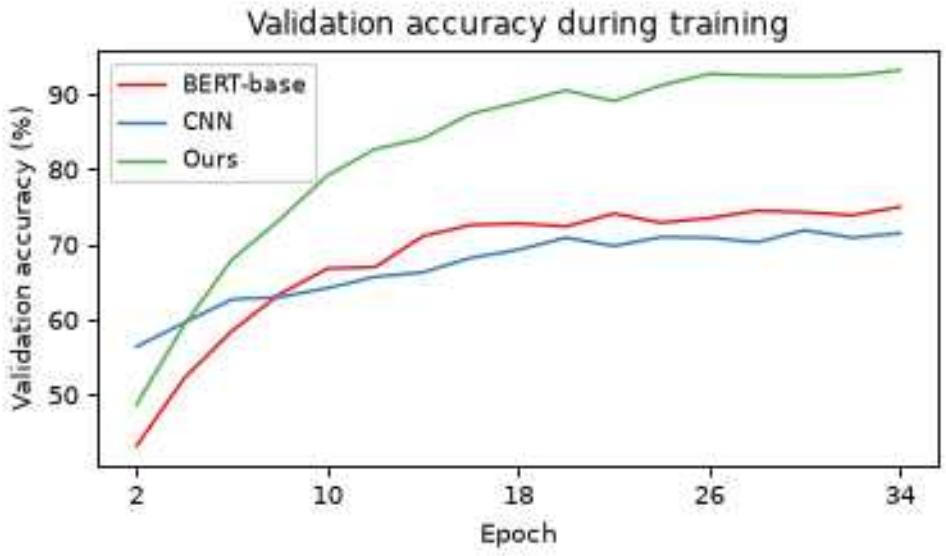



Figure 2: Validation accuracy during training for the models in this experiment.

At epoch 6, BERT-base achieves 58.5% validation accuracy (Figure 2). As shown in Figure 2, CNN reaches 65.2% validation accuracy at epoch 14. As shown in Figure 2, Ours outperforms CNN by 20.5 percentage

points at epoch 30.

Figure 3 shows the f1 score by dataset.

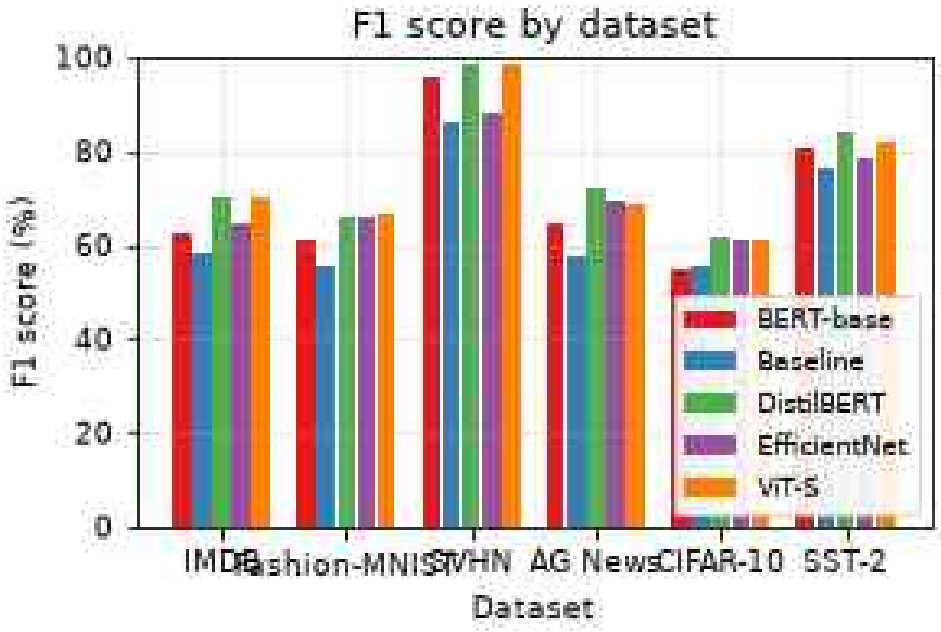



Figure 3: F1 score by dataset for the models in this experiment.

DistilBERT outperforms BERT-base by 17.0 percentage points on CIFAR-10 (Figure 3). Baseline achieves 136.2% F1 score on SVHN (Figure 3). On CIFAR-10, ViT-S is 7.5 points better than Baseline. ViT-S outperforms BERT-base by 24.2 percentage points on AG News (Figure 3).

Figure 4 shows the f1 score during training.

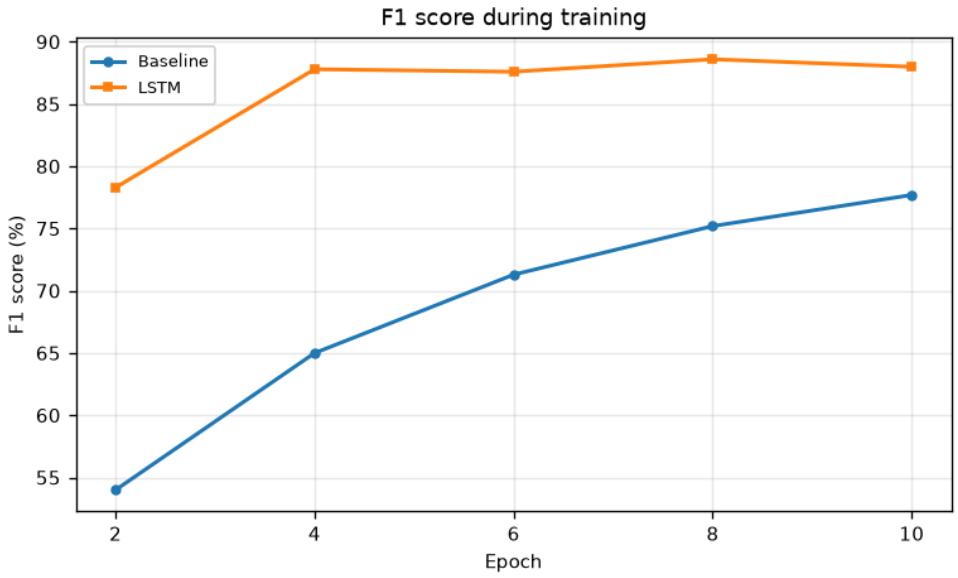



Figure 4: F1 score during training for the models in this experiment.

At epoch 2, Baseline achieves 78.3% F1 score. At epoch 2, Baseline achieves 54.0% F1 score. From epoch 6 to epoch 8, the F1 score of Baseline improves by a relative 5.5% (Figure 4).

## 3 Discussion

Logistic Regression reaches 65.6% accuracy on IMDB. We note that the training was stable for all models.